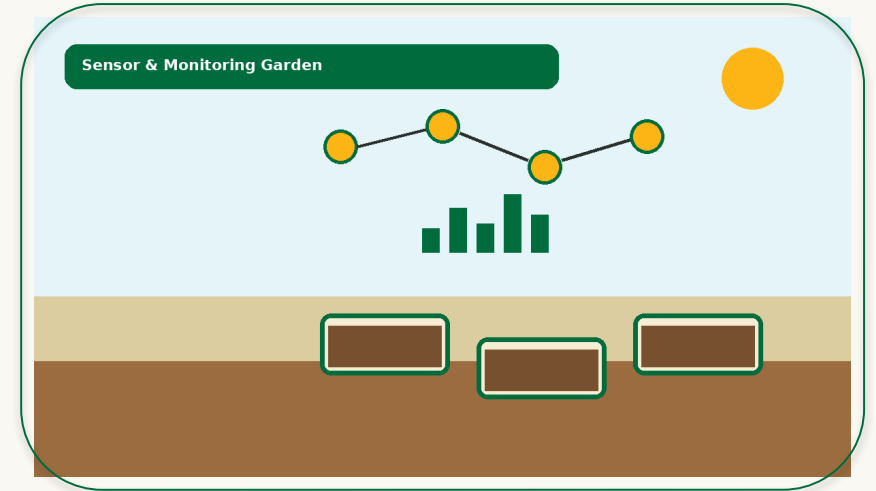


# Sensor & Monitoring Garden

## Agricultural Instrumentation and Data

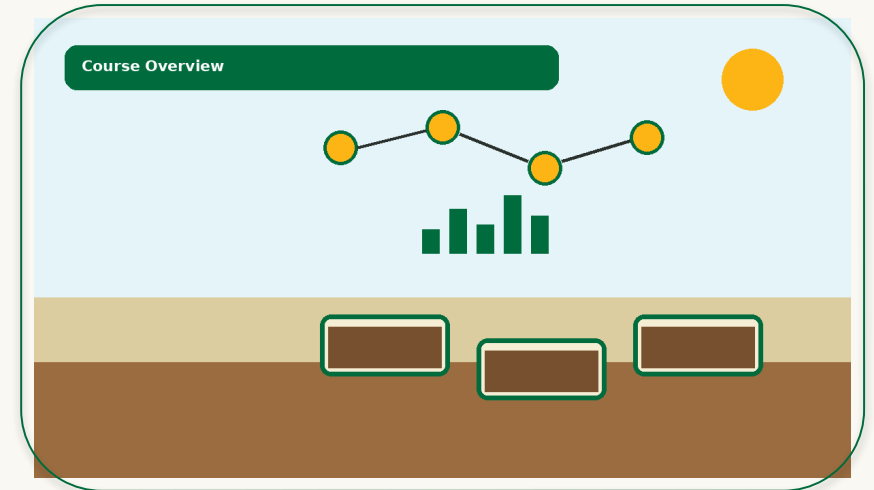
- LIFEWS AgriShine™ GrowMeal™ SSS 1-3 Teacher Training
- Audience: SSS 1-3 Agricultural Science, Biology, Mathematics and technical/STEM educators
- Format: Self-paced online course + supervised field/laboratory practical
- Recommended duration: 4 hours online + 2-3 hours practical
- SSS learning progression: design -> instrument -> model -> evaluate -> optimize -> innovate



# Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the system at senior-secondary technical depth; teach all ten modules accurately; supervise design, instrumentation, measurement and experimental work; maintain defensible technical records; guide quantitative analysis; distinguish measurement from inference; connect evidence to management, engineering and research decisions.
- Core materials: teacher guide, system drawings, approved tools, low-voltage educational hardware where relevant, measuring instruments, meters/sensors, data sheets/logger, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe technical practical, and submit an engineering design, monitoring plan, experimental protocol or systems-analysis report appropriate to the course.



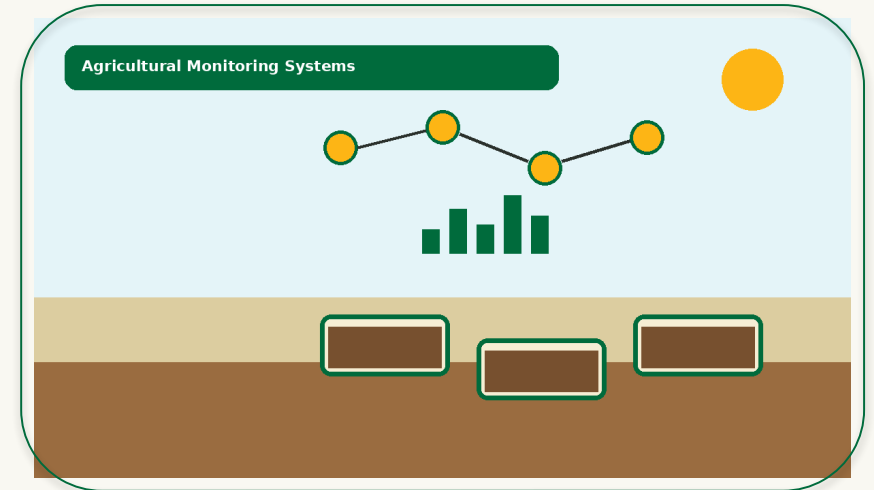
## TEACHER PRACTICE

- Training sequence: DEFINE -> DESIGN -> IMPLEMENT -> MEASURE -> ANALYZE -> EVALUATE -> IMPROVE.

# Agricultural Monitoring Systems

Teacher technical knowledge, laboratory/field practice and student supervision

- Define a monitoring system as sensors, data acquisition/logging, storage, visualization and interpretation working as a chain.
- Map each measured variable to a management or research purpose.
- Distinguish continuous, periodic and event-based monitoring.
- Identify power, communication and data-storage requirements conceptually.
- Design monitoring around questions rather than collecting data without purpose.



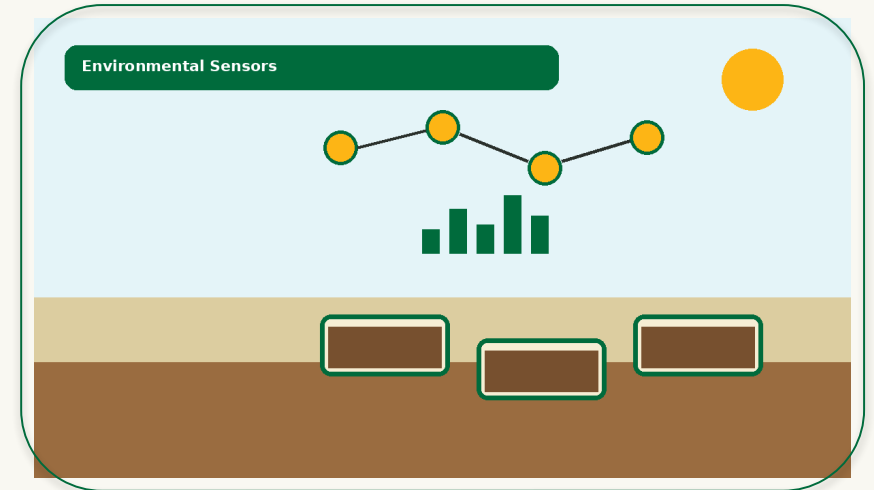
## TEACHER PRACTICE

- Teacher task: design a monitoring architecture for one school production system.

# Environmental Sensors

Teacher technical knowledge, laboratory/field practice and student supervision

- Use appropriate sensors for air temperature, relative humidity, rainfall, radiation/light or other installed variables.
- Place sensors according to exposure and shielding requirements.
- Record height, location, units and sampling interval.
- Recognize sensor response time and environmental interference.
- Check questionable readings against a reference or nearby observation.



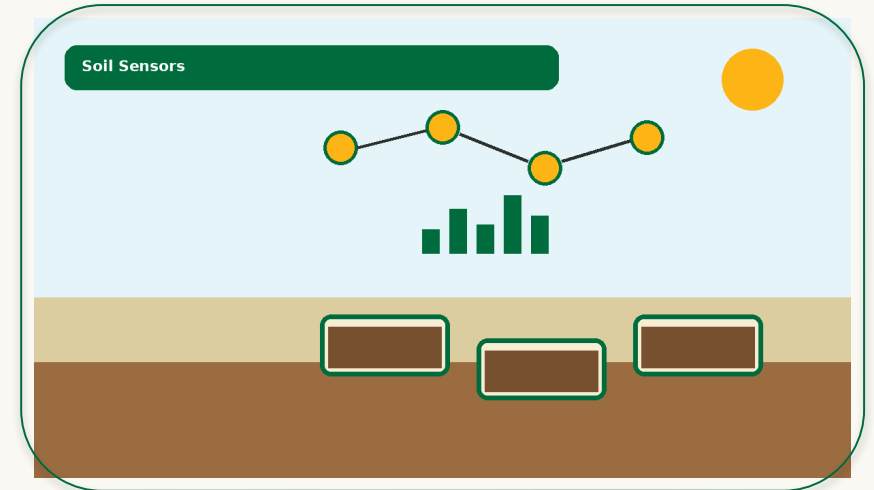
## TEACHER PRACTICE

- Prepare an environmental-sensor installation plan.

# Soil Sensors

Teacher technical knowledge, laboratory/field practice and student supervision

- Identify soil-moisture, temperature or other soil sensors used by the school.
- Select depth and position according to root zone and monitoring question.
- Install with good soil contact where required.
- Use multiple locations to address spatial variability.
- Document soil type, depth and nearby irrigation features.



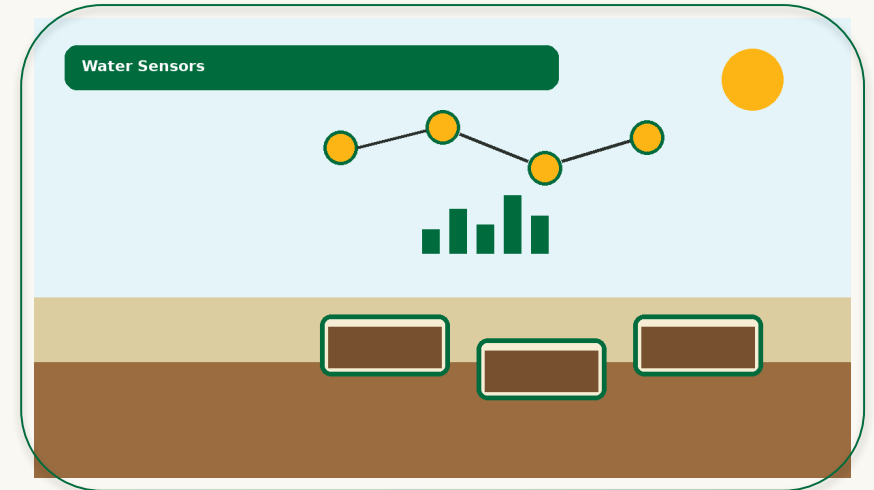
## TEACHER PRACTICE

- Design a soil-sensor sampling network.

# Water Sensors

Teacher technical knowledge, laboratory/field practice and student supervision

- Monitor variables such as flow, level, pressure, temperature or water quality where equipment is installed.
- Distinguish instantaneous flow from cumulative volume.
- Position sensors according to manufacturer/system requirements.
- Record operating state when readings are taken.
- Keep students away from electrical, pressurized and chemical-contact service work.



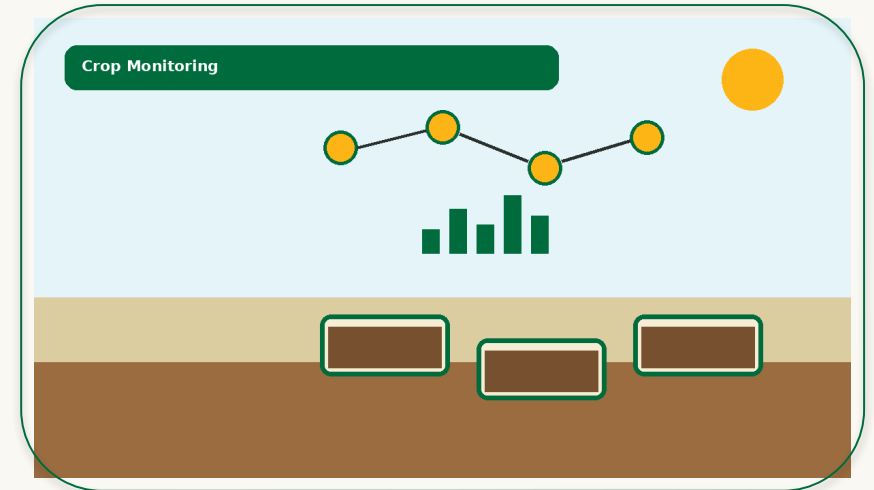
## TEACHER PRACTICE

- Create a water-monitoring point map.

# Crop Monitoring

Teacher technical knowledge, laboratory/field practice and student supervision

- Combine direct crop measurements with imaging or simple sensor observations where available.
- Define traits such as canopy size, height, phenology, leaf condition or yield.
- Use consistent observation positions and timing.
- Avoid treating visual indices as direct substitutes for yield or crop health without validation.
- Link crop observations to environmental and management records.



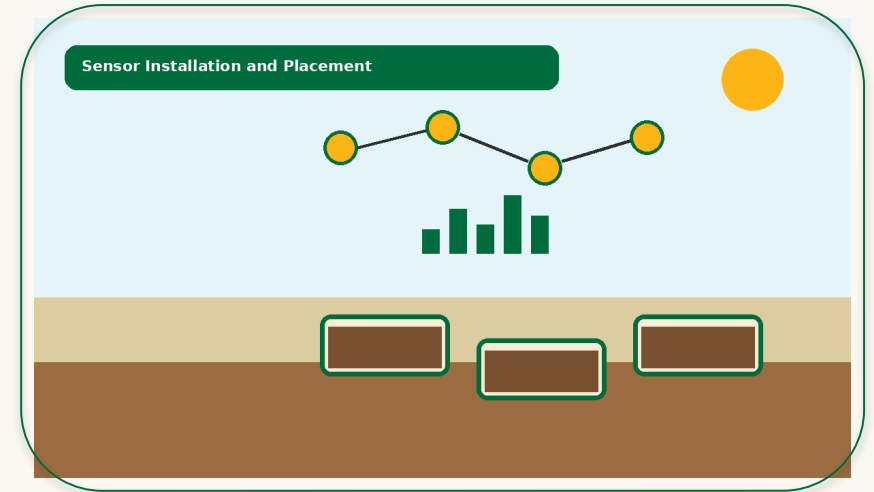
## TEACHER PRACTICE

- Prepare a crop-monitoring protocol using manual plus digital observations.

# Sensor Installation and Placement

Teacher technical knowledge, laboratory/field practice and student supervision

- Define the measurement question before selecting placement.
- Avoid biased positions unless the bias is intentionally studied.
- Document coordinates/plot, depth, height, orientation and nearby structures.
- Protect sensors and cables from tools, water intrusion and trampling.
- Reassess placement as canopy and system conditions change.



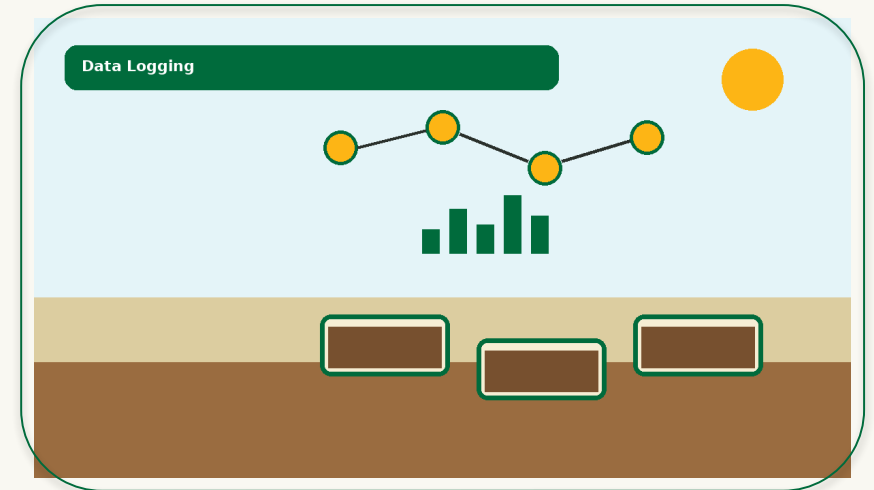
## TEACHER PRACTICE

- Produce an installation record for each sensor.

# Data Logging

Teacher technical knowledge, laboratory/field practice and student supervision

- Set sensor IDs, timestamps, units and sampling intervals consistently.
- Synchronize clocks where multiple devices are compared.
- Identify missing values, communication gaps and impossible readings.
- Preserve raw data before cleaning.
- Maintain backups and a data dictionary.



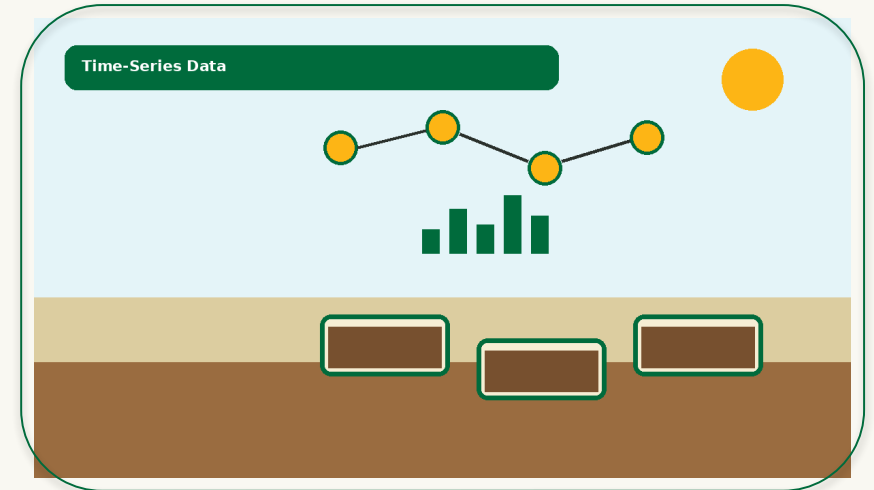
## TEACHER PRACTICE

- Create a monitoring data dictionary and logging SOP.

# Time-Series Data

Teacher technical knowledge, laboratory/field practice and student supervision

- Plot measurements against time to reveal cycles, trends, peaks and abrupt changes.
- Relate time-series patterns to irrigation, rainfall, system operation and crop stages.
- Use consistent time zones and intervals.
- Recognize autocorrelation conceptually: nearby measurements may not be independent.
- Avoid interpreting short-term fluctuations without context.



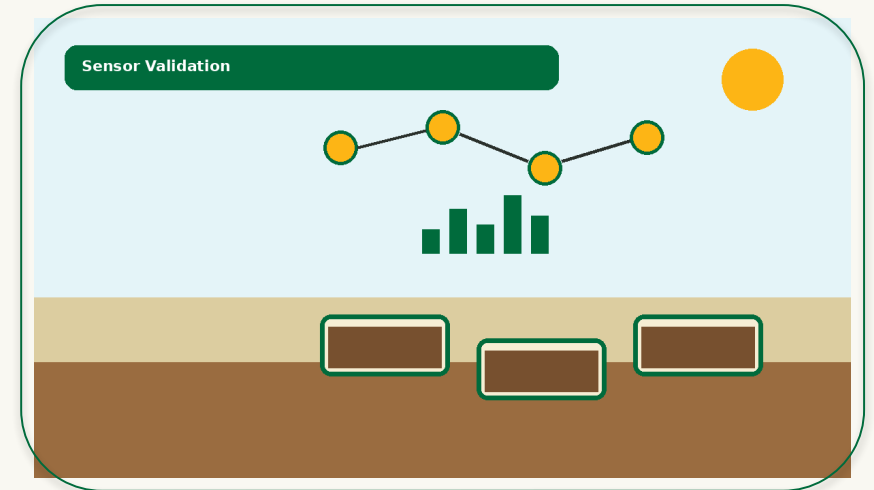
## TEACHER PRACTICE

- Prepare and interpret a 7-day sensor time series.

# Sensor Validation

Teacher technical knowledge, laboratory/field practice and student supervision

- Compare sensor readings with a trusted reference, manual method or second instrument where appropriate.
- Use paired measurements across a useful range, not one comparison point.
- Calculate simple differences, bias or agreement summaries.
- Check calibration, placement and units when disagreement occurs.
- Document validation date and outcome.



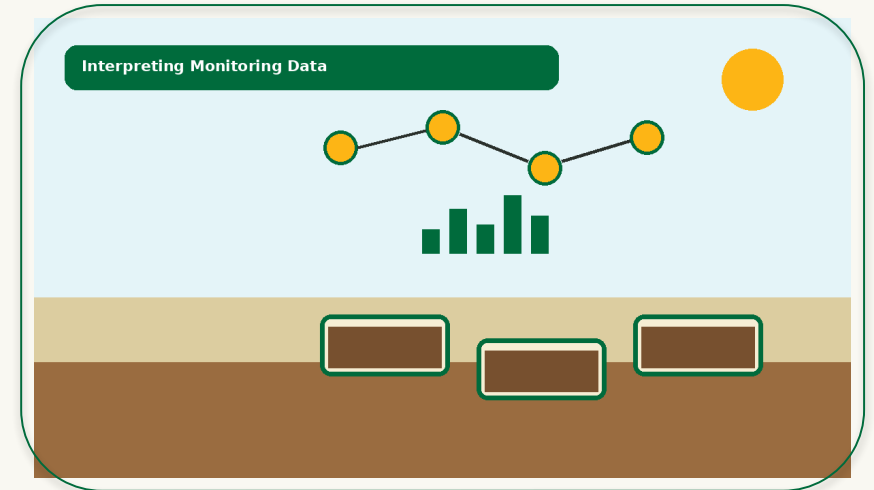
## TEACHER PRACTICE

- Design a sensor-validation exercise.

# Interpreting Monitoring Data

Teacher technical knowledge, laboratory/field practice and student supervision

- Integrate environmental, soil, water and crop data with management records.
- Identify relationships and anomalies requiring verification.
- Distinguish correlation, temporal coincidence and causal evidence.
- Use thresholds only when technically justified and locally relevant.
- Translate findings into a monitoring or management recommendation with stated uncertainty.



## TEACHER PRACTICE

- Completion task: produce a monitoring-data interpretation report.